Hi Joshua,

Attached is my Assignment 7 (Complete AI Dev Pipeline) for DUMMIES.

Shipping V1, is also playable: https://lawrencelek.github.io/ELVTR-Multiagent-Submission-7/p11/
Pipeline repo: https://github.com/lawrencelek/ELVTR-Multiagent-Submission-7
Pipeline run video: https://lawrencelek.github.io/ELVTR-Multiagent-Submission-7/docs/pipeline_run.mp4

Thanks,
Lawrence

DUMMIES — Assignment 7: Complete AI Dev Pipeline

11 November 2026 · Lawrence Lek

An agent pipeline writes Map 1's level plan; the browser game loads it unchanged. Shipped run v3g cost $0.9660. Play: https://lawrencelek.github.io/ELVTR-Multiagent-Submission-7/

Student & Game Overview

Field	Answer
Student Name	Lawrence Lek
Capstone Game Title	DUMMIES
Game Concept Brief	A top-down browser game. You are a crash-test dummy in a smart city. You score by getting hit by a self-driving car that learns. Each crash teaches it a new sensor, which closes the trick you used. The pipeline designs Story Map 1: nine tiles, the car's learning order, each tile's goal, card text and hint.

Deliverable 1: Playable Link

Playable Game Link: https://lawrencelek.github.io/ELVTR-Multiagent-Submission-7/

- **Host:** GitHub Pages, one static page. No install, no login, no download. Desktop browser with a keyboard.
- **No instructions needed.** Every screen names its keys:
  - Title: SPACE START.
  - Menus: ARROWS CHOOSE · SPACE SELECT.
  - Name entry: ENTER OK.
  - Tile card: the goal. "1 damaging hit by the red car. Open road: walk into its path; let it hit you."
  - In play: WASD MOVE.
- **Checked 11 Nov 2026:** a scripted first-time player reached play in about 10 key presses, under a minute. No page errors.

Deliverable 2: Pipeline Source Code & Engine Integration

Pipeline Repository Link: https://github.com/lawrencelek/ELVTR-Multiagent-Submission-7

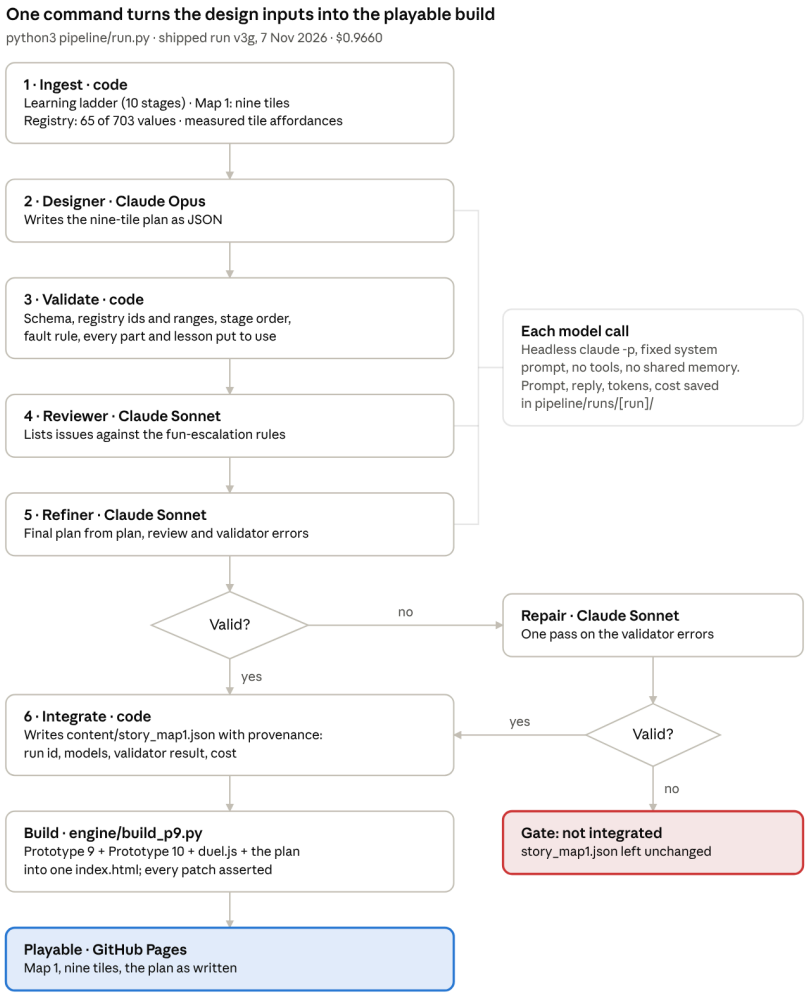
Pipeline Run Video Link: https://lawrencelek.github.io/ELVTR-Multiagent-Submission-7/docs/pipeline_run.mp4 (2 min 48 s)

- 0:00–0:56: the v1, v2 and v3g runs, replayed from their own logs with recorded timestamps. Model waits are shortened and labelled.
- The shipped run, v3g, ends: `wrote content/story_map1.json`, `wrote index.html`, run cost $0.9660.
- Then the plan it wrote: nine tiles, goals, titles, provenance.
- 0:56 on: the game playing that plan, tile 1 onward.

Integration Breakdown

Target Game Engine: a custom HTML5 Canvas 2D engine in plain JavaScript, running in the browser. Prototype 9's game with Prototype 10's car AI. No framework.

Automated Flow Description: one command, `python3 pipeline/run.py`. Agents write a JSON plan. Code validates it. The integrate step writes `content/story_map1.json`, and `engine/build_p9.py` bundles it into `index.html`. The engine reads the file as written: no hand edits, no reformatting.



Plan field	What the engine does with it
`stage`	Loads that stage of the car's learning ladder: start rung, top rung, sensors
`params`	Applies registry overrides; ids and ranges validated first
`objective`	Sets the hits needed to clear the tile; shown in the HUD
`title`, `tale`, `hint`, `newIdea`	Fill the tile card, the objective line and the NEW line
`provenance`	Records run id, models, validator result, cost

- **Gate:** a plan that fails validation after one repair is not integrated.
- **One manual step remains:** publishing. A `git push` puts the build on GitHub Pages.

Deliverable 3: Pipeline Audit & Cost Analysis (1 page)

Pipeline Production & Functionality

What did the pipeline produce? Map 1's level plan, `content/story_map1.json` (run v3g). All nine tiles of the build play from it:

- the car's learning order: one new sensor or dummy part per tile;
- each tile's goal, shown in the HUD: "LET IT HIT YOU", "DROP INTO ITS PATH", "DUCK, THEN DASH";
- nine tile titles and tales: "The welcome wall": "Fresh off the factory floor. Your first colleague is red, keen and blind.";
- nine hints that name the tile's cover: "Hide behind the kiosk at the kerb; step into its path late.";
- two tuning overrides: A1 `duel.redReturnSeconds` 3; C1 `ui.lungeGuide` 1.

The file carries its provenance: run id, models, validator result, cost. Per-tile list: Appendix A.

What manual steps remain?

- Publishing: a `git push` to GitHub Pages.
- Prototypes 9 and 10 were copied into the repo by hand (md5s in `VENDOR.md`).
- Difficulty: headless bot probes are run and read by hand.
- Fun: a person plays and judges.

What would it take to eliminate them?

- One GitHub Action on push: run the pipeline, test the nine tiles, deploy Pages, render the video.
- `pipeline/vendor.py`: fetch a pinned Submission 6B commit and check the md5s.
- Run the probes (`engine/p9_probe.js`) inside ingest. Feed hits a minute to the designer as a measured fact.
- Fun stays human. A recorded bot session per tile in CI shortens the review.

Architectural Reflection

Current Architectural Decision to Change: joining the two prototypes by patching. The red car's AI (Prototype 10) enters Prototype 9's traffic at runtime. `engine/duel.js` swaps the tile's sim for a subclass, and the build string-patches `game.js`. It works, but it depends on seven internal names: `admit`, `meets`, `level.statics`, `manSpeed`, `watchField`, `spawnSpeed`, `searchStep`. Any upstream rename breaks a patch.

Specific Alternative: a driver interface in the engine. Prototype 9's traffic sim calls `driver.perceive(world)`, `driver.decide(dt)` and `driver.onContact(hit)` for every vehicle. The fleet keeps the default driver. The red car gets Prototype 10's brain as a driver module, configured by the plan's `stage`. No patches: the build only bundles.

Cost Analysis

Total Actual Run Cost: $0.9660. The shipped content run, v3g (`pipeline/runs/20261107-182619-v3/costs.json`): 3 model calls, 122,091 input and 33,034 output tokens, 4 min 15 s.

Most Expensive Pipeline Step: the designer, on Claude Opus: $0.5396, 56% of the run. It writes the longest reply (15,488 output tokens) at Opus rates. Reviewer (Sonnet) $0.2406; refiner (Sonnet) $0.1858.

Solo/Small-Team Sustainability: yes. A content run costs $0.52–$0.97. Three maps × 10 iterations cost $16–$29. Every agent tool used on Map 1 cost $28.68 recorded. The game makes no model calls at runtime, so players cost nothing.

Mid-Project Cost-Reduction Change

	Before (v1)	After (v2)
Strategy/Prompting Approach	Full registry (703 values) in every prompt. Opus for designer, reviewer and refiner	Registry filtered to the 65 values the ladder uses. Opus designer only; Sonnet reviewer, refiner, repair
Token / API Cost	461,915 input tokens · $3.9092 a run	126,199 input tokens · $0.7343 a run (−81%)

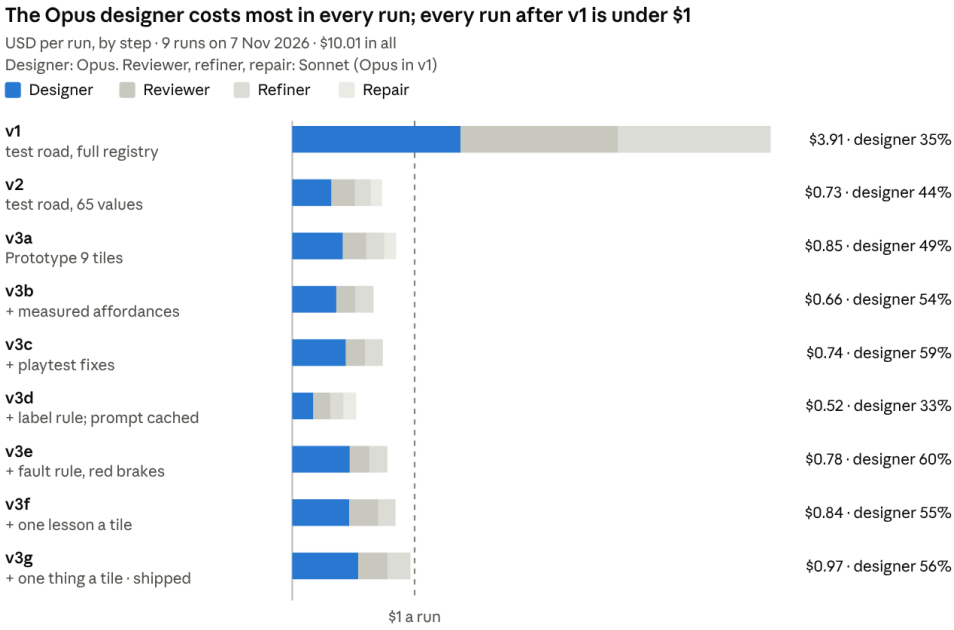
Appendix: evidence behind the audit

A. The nine tiles the pipeline planned

Tile	Stage	Goal in the HUD	Card title	One new thing
A1	10.2	LET IT HIT YOU × 1	The welcome wall	Car learns the forward camera
B1	10.3	STEP OUT AT THE KERB × 1	The podium bridge	Dummy earns Battery 1 (dash)
C1	10.3	DASH INTO ITS PATH × 2	The LED drum	Car learns side sensors
C2	10.4	DROP INTO ITS PATH × 1	The portal tower	Dummy earns Hands 1 (cone)
B2	10.4	CONE-BAITED HIT × 1	The monument	Dummy earns Legs 1 (jump)
A2	10.4	JUMP INTO ITS PATH × 1	The neon shops	Car learns roof lidar
A3	10.5	DASH FROM COVER × 1	The stacked cubes	Dummy earns Super Knees (duck)
B3	10.5	DUCK, THEN DASH × 1	The LED canyon	Car learns street cameras
C3	10.6	UNSEEN DASH × 1	The LED gateway	Car learns road items

- **Checks:** validator 0 errors. The reviewer asked for a revision (tiles 7–8 needed tricks their stage cannot offer); the refiner fixed it; validator 0 errors.
- **In the build, re-tested 11 Nov 2026:** `engine/p9_tiles_test.js`: 9 of 9 tiles start with the plan's stage, rung, goal and cover; no page errors.

B. Cost of every content run



Source: `pipeline/runs/*/costs.json` · 9 content runs, 30 model calls, 7 Nov 2026.

- **Content pipeline total:** $10.0053 over 9 recorded runs, 30 calls. Two runs stopped at the gate left no record (about $1 each).
- **Cache:** run back to back, the designer's prompt stays cached. v3d's designer read 32,143 tokens from cache: $0.17, against $0.44 in v3c.

C. Every agent tool used on Map 1

Agent (script)	Runs	Recorded cost
Content pipeline (`run.py`)	9	$10.01
Adversarial QA (`qa_attack.py`)	1 (3 calls)	$4.55
Car design (`car_design.py`)	2 (8 rounds)	$4.88; 4 rounds unrecorded, about $5
Upgrade-screen redesign (`ui_upgrade.py`)	4	$4.46
UI review (`ui_review.py`)	2	$3.85
Puzzle pressure test (`pressure_test.py`)	1	$0.94
Total	**19**	**$28.68 recorded; about $36 with unrecorded calls**

- **Lost records, fixed 11 Nov:** resumed runs kept call costs only in memory. `pipeline/llm.py` now appends every call to `calls.jsonl` in its run folder. Testing that fix spent $0.16.
- **Not metered here:** the Claude Code sessions that built the pipeline and engine (subscription).

D. The v1 to v2 reduction in detail

	v1	v2	Change
Registry sent	703 entries	65 entries	−91%
Agent input	296,724 characters	27,432 characters	−91%
Input tokens	461,915	126,199	−73%
Output tokens	18,864	20,959	+11%
Cost a run	$3.9092	$0.7343	−81%

- **Where it came from:** the filter alone cut the Opus designer by 77% ($1.3757 to $0.3205). The Sonnet swap cut reviewer and refiner by 84%.
- **Quality held:** v2 passed the same validator, after one repair. v3b, with Prototype 9's tiles and measured facts, cost $0.66 and needed no repair.

E. Beyond the submission: Shipping V1

The puzzle version of Map 1: https://lawrencelek.github.io/ELVTR-Multiagent-Submission-7/p11/. Its puzzle data is written by hand. Agent output inside it:

- **Hero cars:** `engine/hero_cars.js`, written by the car design agent over 4 self-review rounds. Bundled as written, except one sentence. Three cars, 21 parts.
- **Upgrade screen:** from the adversarial UI redesign agent's spec and three review rounds.
- **QA:** an adversarial QA agent played the nine puzzles through a virtual pad: 87 attack plans, 15 bug reports (GDD Edit 118).

Rubric map

Criterion	Points	Where it is met
Playable Link	2.0	Deliverable 1: one URL; every screen names its keys; play in under a minute
Pipeline-to-Game Connection	3.0	Video: the run log writes `story_map1.json`, then the game plays it. Deliverable 3: what the pipeline produced. Appendix A: the nine tiles; provenance in the file
Engine Integration	2.0	Deliverable 2: Automated Flow Description and field table. One documented manual step: publishing
Cost Analysis	2.0	Deliverable 3: Cost Analysis, from the actual run. Appendix B and C
Pipeline Audit	1.0	Deliverable 3, one page, every field of the brief. Detail in the Appendix

Sources

- Playable build · Shipping V1, puzzles
- Repository · README
- Pipeline-run video · Shipping V1 runthrough video (Appendix E)
- The plan, `content/story_map1.json` · run folders, `pipeline/runs/`
- Pipeline audit, `docs/02_Pipeline_Audit.md` · Prototype 9 × 10 integration analysis
- GDD: Edit 111, Submission 7 integration · Edit 118, QA agent · Edit 121, car design agent